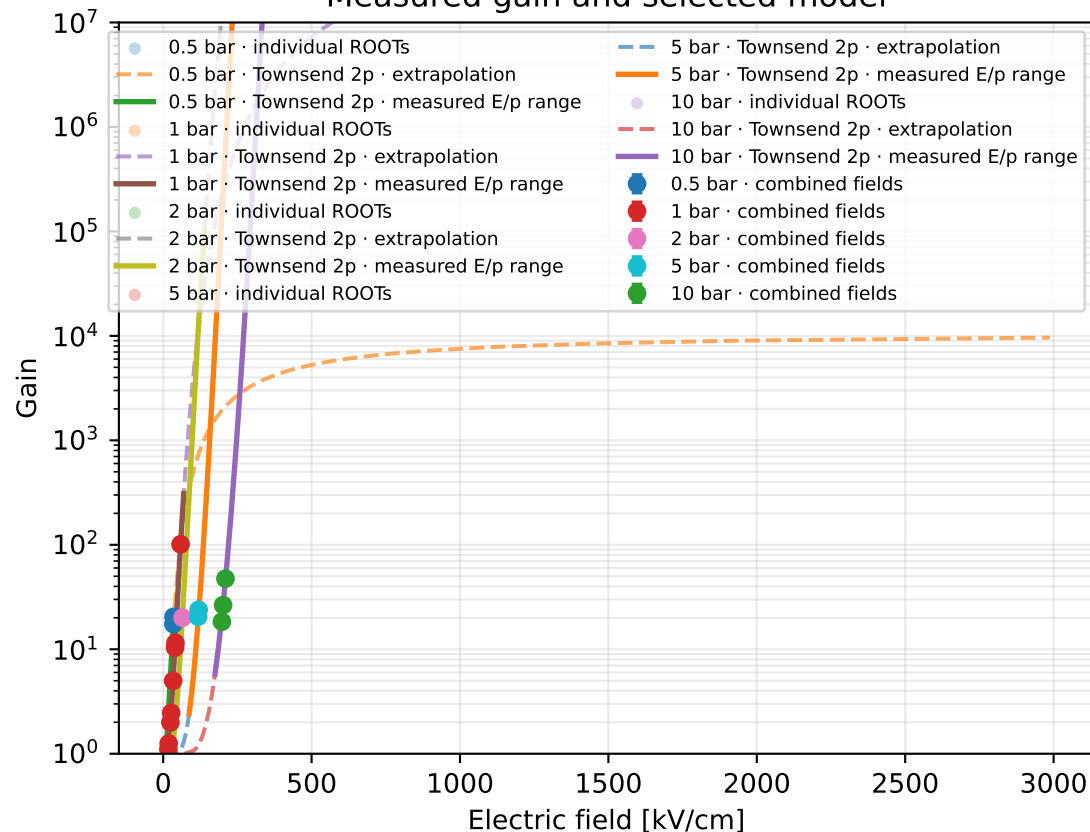
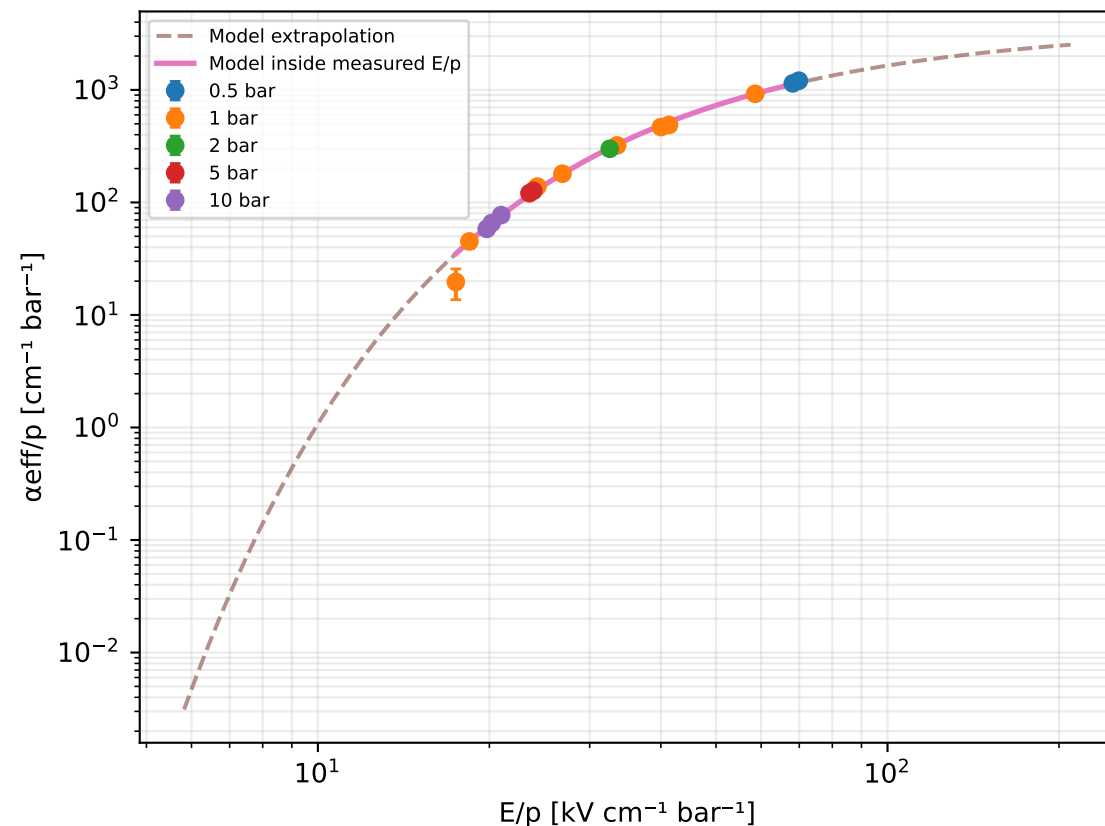


ArCF₄ · Ar 90 % / CF₄10 % · gap 0.05 mm

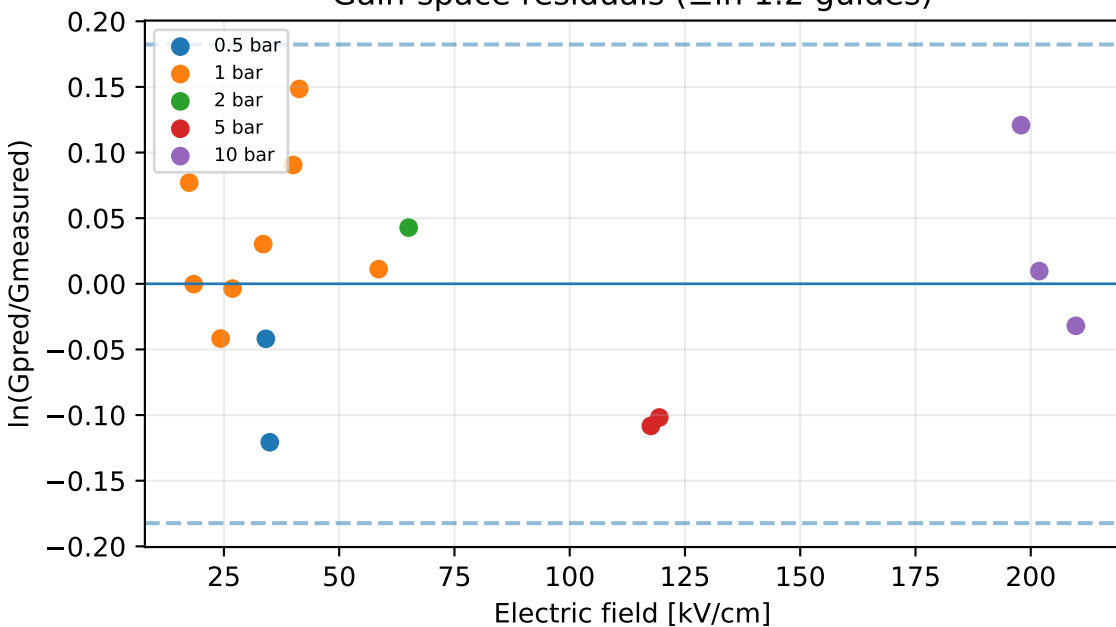
Measured gain and selected model



Reduced Townsend fit

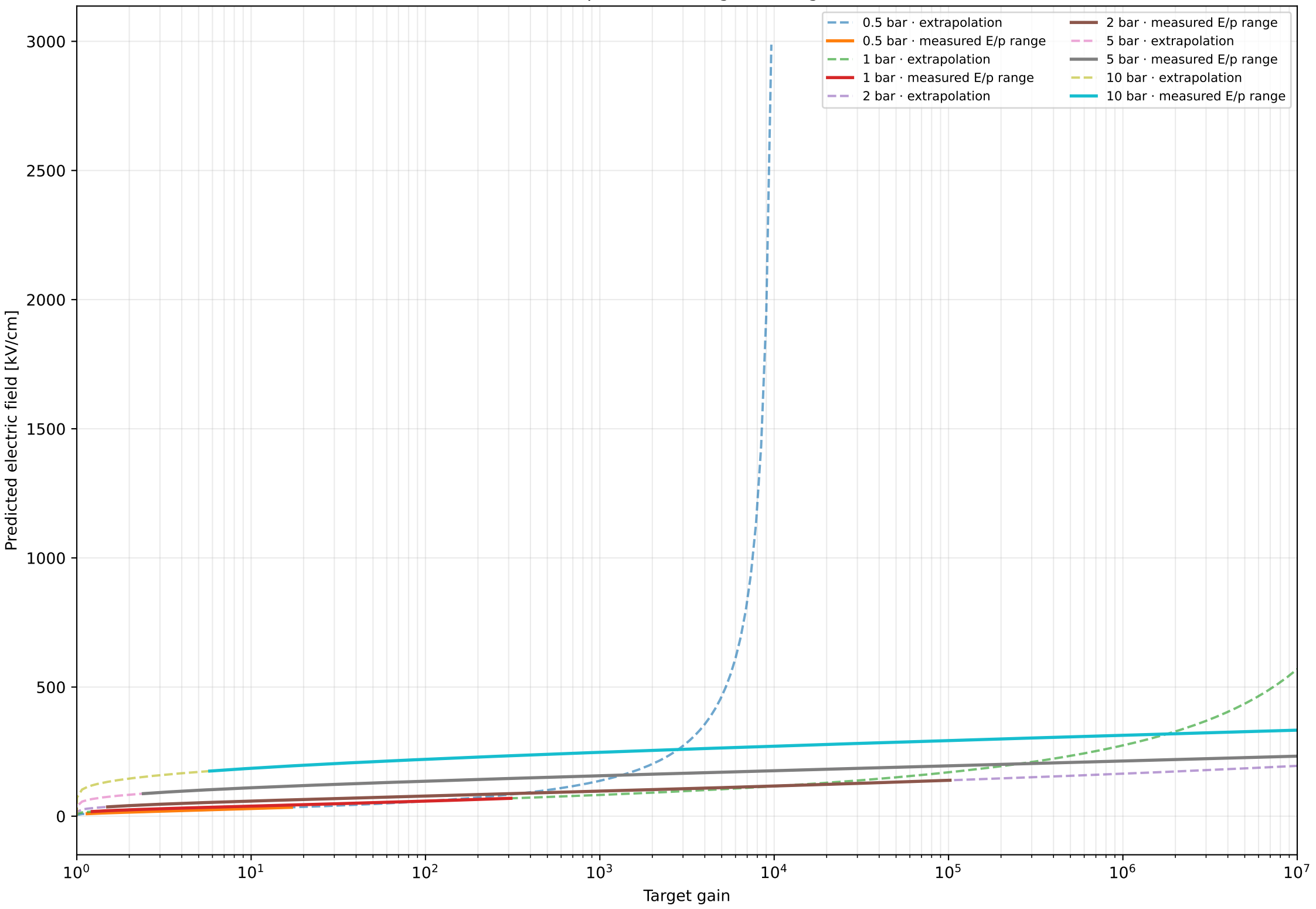


Gain-space residuals ($\pm \ln 1.2$ guides)



ROOT runs: 31
 Unique physical points: 16
 Pressures: 0.5, 1, 2, 5, 10 bar
 Selected model: Townsend 2p
 Status: accepted
 Training RMSE: $\times 1.160$
 Cross-validation median: $\times 1.059$
 Cross-validation worst: $\times 1.177$
 Reduced-field range: 17.47–69.88 kV cm⁻¹ bar⁻¹
 A=3719.5, B=81.461, m=0, n=1
 Covariance condition: 80.4

Field extrapolation from gain 1 to gain 10^7



p [bar]	E [kV/cm]	E/p	Gain	σ Gain	α_{eff} [cm ⁻¹]	npe	runs
0.5	34.0959	68.1917	17.42347	0.522	571.564	392	1
0.5	34.9407	69.8814	20.47343	0.614	603.826	414	1
1	17.4658	17.4658	1.103245	0.0331	19.6511	339	1
1	18.451	18.451	1.252551	0.0375	45.0365	392	1
1	24.2833	24.2833	1.99604	0.0599	138.233	505	1
1	26.8704	26.8704	2.46139	0.0738	180.145	518	1
1	33.5164	33.5164	4.983957	0.149	321.245	561	1
1	40.0215	40.0215	10.36832	0.311	467.751	505	1
1	41.3391	41.3391	11.5148	0.345	488.727	608	1
1	58.5608	58.5608	101.1033	3.63	923.229	300	1
2	65.0719	32.536	20.06466	0.602	599.792	696	1
5	117.597	23.5194	20.5	0.615	604.085	980	1
5	119.432	23.8865	23.88411	0.716	634.643	1139	1
10	197.894	19.7894	18.37747	0.551	582.225	1571	1
10	201.841	20.1841	26.48101	0.794	655.286	1422	1
10	209.768	20.9768	47.44259	0.356	771.904	22043	16

Progressive model selection

The higher-order model is selected only when new points improve out-of-sample prediction.

Candidate	k	AICc	CV median	CV worst	usable	warnings
Townsend 2p	2	-56.2	×1.06	×1.18	yes	—
Townsend 3p	3	-51.6	×1.06	×1.25	yes	—
Townsend 4p	4	-60.2	×1.02	×1.11	yes	—